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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/690,076	03/07/2024	Shota ISHII	2024-0265A	6639

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EXAMINER
RHIM, WOO CHUL

ART UNIT	PAPER NUMBER
2676	

NOTIFICATION DATE	DELIVERY MODE
06/02/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/690,076	Applicant(s) ISHII et al.	
	Examiner WOO RHIM	Art Unit 2676	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 05/26/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-9 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/07/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The information disclosure statement (IDS) submitted on 03/17/2026 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Response to Amendment

3. Submission dated 05/26/2026 amends the specification and claims 1-2 and 5-9. Claims 1-9 are pending.

4. In view of the amendment to the specification, the previously set forth objections to the specification have been withdrawn, and in view of the amendment to the claims, the previously set forth claim objections and 112(b) rejections have been withdrawn.

Response to Argument

5. On page 8 of the submission, the applicant states that the system claims have been amended to not invoke the 112(f) interpretation. The examiner points out that adding the phrase “configured to” does not take the respective limitations out of the 112(f) interpretation because the limitations still use a generic place holder, e.g., “unit”, that does not denote any structure **(see MPEP 2181(I)).**

The examiner suggests, if supported, replacing the generic placeholder with a structural term such as a processor, a circuit, or a circuitry.

6. On pages 8-10 of the submission, the applicant argues that Shota does not teach or suggest "estimating a coordinate transformation parameter to transform a first coordinate system corresponding to a three-dimensional Computer Aided Design model into a second coordinate system being a two-dimensional coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image, based on the shape recognized by the identification shape unit and on the three-dimensional Computer Aided Design model of the inspection target object." The examiner disagrees.

First, as discussed before in the previous office action and again in the below rejection, Shota as applied implicitly teaches or at least suggests the quoted limitations because the cited portion of Shota teaches extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified. The internal parameters used in extracting the 2D simulated image corresponds to the claimed coordinate transformation parameters because 3D CAD model is transformed into the 2D simulated image. The examiner notes that the coordinate system of the extracted 2D simulated image corresponds to the claimed second coordinate system because the extracted 2D simulated image since it corresponds to the 2D real image and contains the same reference parts of the inspection object as the real image.

Second, as also discussed before in the previous office action and again in the below rejection, the limitations are also taught by Sakamoto because the cited portion of Sakamoto teaches estimating the relative position and posture of the camera between the selected frame and the reference frame to transform a set of coordinates corresponding to the 3d model into another set of coordinates corresponding to the viewpoint of the camera that has captured the reference frame based on the 3d model and the 3d shape data therein **(see, e.g., pars. 156-165 and FIGS. 2 and 5-7 of Sakamoto).**

For the aforementioned reasons, the examiner disagrees with the applicant's arguments and finds the applicant's arguments unpersuasive.

7. On page 10 of the submission, the applicant argues that Sakamoto does not teach or suggest "extracting a two-dimensional simulated image corresponding to a two- dimensional photographed image from a three-dimensional Computer Aided Design model after the viewpoint information has been modified, as in independent claim 1." The examiner disagrees.

First, the examiner points out that the examiner did not rely on Sakamoto to teach the quoted limitation. Instead, the examiner relied on Shota. As such, the applicant's argument moot.

Second, Shota as applied taches the quoted limitation because it teaches extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6 of Shota).**

For the aforementioned reasons, the examiner disagrees with the applicant's arguments and finds the applicant's arguments unpersuasive.

Claim Interpretation

8. The claim interpretation under 35 U.S.C. 112(f) is maintained in view of the amendments to the claims for the reasons provided in the item #5 of the Response to Argument section.

9. As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

10. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language

without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are:

"identification shape unit" in claim 1;

"defect detection unit" in claim 1;

"coordinate transformation parameter estimation unit" in claim 1;

"three-dimensional CAD model position change unit" in claim 1;

"two-dimensional simulated image extraction unit" in claim 1;

"depiction unit" in claim 1;

"report creation unit" in claim 6; and

"coordinate transformation parameter correction unit" in claim 7.

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 103

11. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention

> and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

12. Claim(s) 1-6 and 8-9 is/are rejected under 35 U.S.C. 103 as being unpatentable over JP patent application publication no. 2021-021669 to Shota et al. (hereinafter Shota) in view of us patent application publication no. 2024/0193852 to Sakamoto.

13. For claims 1 and 8, Shota as applied teaches an inspection assistance system comprising:

an identification shape unit configured to recognize a shape of an inspection target object included in a two-dimensional photographed image based on the two-dimensional photographed image obtained by capturing the inspection target object with an imaging device **(see, e.g., par. 47 and FIG. 1, which teach identifying a shape of the inspection object in a 2D real image acquired by an image acquisition unit)**;

a defect detection unit configured to detect a defect of the inspection target object included in the two-dimensional photographed image **(see, e.g., par. 56 and FIG. 1, which teach detecting a defect of the object in the 2Dreal image)**;

a coordinate transformation parameter estimation unit configured to estimate a coordinate transformation parameter to transform a first coordinate system corresponding to a three-dimensional Computer Aided Design model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image based on the shape recognized by the identification shape unit and on the three-dimensional Computer Aided Design model of the inspection target object **(see, e.g., par. 57 and FIG. 1, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**, the second coordinate system being a two-dimensional coordinate system **(see, e.g., par. 57 and FIG. 1, which**

teach extracting a 2D simulated image; the examiner finds the coordinate system of the simulated to be 2D coordinate system);

a three-dimensional Computer Aided Design model position change unit configured to modify a position and a direction of viewpoint information of the three-dimensional Computer Aided Design model of the inspection target object by using the coordinate transformation parameter **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**;

a two-dimensional simulated image extraction unit configured to extract a two-dimensional simulated image corresponding to the two-dimensional photographed image from the three-dimensional Computer Aided Design model after the viewpoint information is modified **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**; and

a depiction unit configured to create a defect image by fitting the two-dimensional photographed image including the defect detected by the defect detection unit to the two-dimensional simulated image, and to depict the defect image on the three-dimensional Computer Aided Design model **(see, e.g., pars. 58-59 and 71 and FIGS. 1 and 4, which teach creating the adjusted defect image by adjusting the 2D defect image showing the defect detected by the detection unit to fit the 2D simulated image, and depicting the adjusted defect image on the 3D CD model)**.

Shota teaches extracting, from the 3d CAD model, a simulated image that corresponds to the real image and contains the reference parts of the real image **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6)**. The examiner believes that the cited portion of Shota may suggest estimating "a coordinate transformation parameter to transform a first coordinate system corresponding to the three-

dimensional CAD model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image based on the shape recognized by the identification shape unit and on a three-dimensional CAD model of the inspection target object" and modifying "a position and a direction of viewpoint information of the three-dimensional CAD model of the inspection target object by using the coordinate transformation parameter."

However, for the interest of compact prosecution, the examiner relies on Sakamoto in the analogous art that explicitly teaches the quoted limitations. Sakamoto teaches:

a coordinate transformation parameter estimation unit to estimate a coordinate transformation parameter to transform a first coordinate system corresponding to the three-dimensional CAD model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image based on the shape recognized by the identification shape unit and on a three-dimensional CAD model of the inspection target object **(see, e.g., pars. 156-165 and FIGS. 2 and 5-7 of Sakamoto, which teach estimating the relative position and posture of the camera between the selected frame and the reference frame to transform a set of coordinates corresponding to the 3d model into another set of coordinates corresponding to the viewpoint of the camera that has captured the reference frame based on the 3d model and the 3d shape data therein)**;

a three-dimensional CAD model position change unit to modify a position and a direction of viewpoint information of the three-dimensional CAD model of the inspection target object by using the coordinate transformation parameter **(see, e.g., pars. 166-167 and FIG. 2 of Sakamoto, which teach modifying a position and posture of viewpoint information of the 3d model by associating the coordinates of the target image with the 3d model based on the position and posture of the camera that acquired the reference image and the condition, i.e., the internal parameter and the distortion correction parameter, for generating the 3d model)**.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the viewpoint of the 3d CAD model as taught by Sakamoto because doing so would yield predictable results of allowing an accurate estimation of the position of the defect when the viewpoints are modified **(see, e.g., par. 166 and MPEP 2143(I)(D)).**

14. For claim 2, Shota as applied teaches that the coordinate transformation parameter estimation unit is configured to specify in advance a plurality of first reference portions of the inspection target object in the two-dimensional photographed image based on the shape recognized by the identification shape unit **(see, e.g., pars. 47-51 and FIGS. 1-2, which teach detecting characteristic parts of the inspection object as the reference parts).**

Shota does not explicitly teach estimating the coordinate transformation parameter such that a plurality of second reference portions registered in advance in the three-dimensional CAD model coincide with the plurality of first reference portions. Sakamoto in the analogous art teaches estimating the relative positions and postures of the camera with respect to the reference frame such that the points in the 3d model coincide with the points of the reference frame **(see, e.g., pars. 164-165 and FIGS. 6-7 of Sakamoto).**

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Shota to use coinciding reference parts as taught by Sakamoto because doing so would yield predictable results of allowing an accurate estimation of the position of the defect when the viewpoints are modified **(see, e.g., par. 166 of Sakamoto and MPEP 2143(I)(D)).**

15. For claim 3, while Shota does not explicitly teach, Sakamoto in the analogous art teaches that the coordinate transformation parameter includes an external parameter for defining a position and a posture of the imaging device in the first coordinate system **(see, e.g., pars. 156-165 and FIGS. 2 and 5-7**

of Sakamoto, which teach estimating the relative position and posture of the camera between the selected frame and the reference frame).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the viewpoint of the 3d CAD model as taught by Sakamoto because doing so would yield predictable results of allowing an accurate estimation of the position of the defect when the viewpoints are modified **(see, e.g., par. 166 of Sakamoto and MPEP 2143(I)(D)).**

16. For claim 4, while Shota does not explicitly teach, Sakamoto in the analogous art teaches that the coordinate transformation parameter further includes an internal parameter relating to the imaging device **(see, e.g., pars. 128 and 166-167 and FIG. 2 of Sakamoto, which teach modifying a position and posture of viewpoint information of the 3d model based the condition, i.e., the internal parameter of the camera)**.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the viewpoint of the 3d CAD model as taught by Sakamoto because doing so would yield predictable results of allowing an accurate estimation of the position of the defect when the viewpoints are modified **(see, e.g., par. 166 of Sakamoto and MPEP 2143(I)(D)).**

17. For claim 5, Shota in view of Sakamoto teaches that the depiction unit adjusts a position and a dimension of the depicted defect image by performing plane transformation of the two-dimensional photographed image including the defect image based on a result of comparing the two-dimensional photographed image with the two-dimensional simulated image **(see, e.g., pars. 58-59 and FIG. 1 of Shota, which teach adjusting the position and dimension of the depicted defect image by performing an affine transformation on the defect image included in the 2D real image based on a result of comparing the two-dimensional real image and the 2D simulated image)**.

18. For claim 6, Shota in view of Sakamoto teaches:

a report creation unit configured to derive a three-dimensional position and dimension of the defect in the inspection target object based on the defect image projected onto the three-dimensional Computer Aided Design model from dimensional data of the three-dimensional Computer Aided Design model, and to create a report including a derivation result of the position and the dimension **(see, e.g., pars. 59-61 and FIG. 1 of Shota, which teach deriving the three-dimensional position and dimension of the defect in the inspection object based on the defect image depicted on the 3D CAD model from the dimension data of the 3D CAD model, and creating a report including the derivation result of the position and the dimension).**

19. For claim 9, Shota as applied teaches a computer-readable storage medium that stores an inspection assistance program that causes a computer **(see, e.g., pars. 33-34, 39, 44, and 92 and FIG. 1)** to execute:

recognizing a shape of an inspection target object included in a two-dimensional photographed image based on the two-dimensional photographed image obtained by capturing the inspection target object with an imaging device **(see, e.g., par. 47 and FIG. 1, which teach identifying a shape of the inspection object in a 2D real image acquired by an image acquisition unit)**;

detecting a defect of the inspection target object included in the two-dimensional photographed image **(see, e.g., par. 56 and FIG. 1, which teach detecting a defect of the object in the 2Dreal image)**;

estimating a coordinate transformation parameter to transform a first coordinate system corresponding to a three-dimensional Computer Aided Design model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional

photographed image, based on the shape and the three-dimensional Computer Aided Design model of the inspection target object **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**, the second coordinate system being a two-dimensional coordinate system **(see, e.g., par. 57 and FIG. 1, which teach extracting a 2D simulated image; the examiner finds the coordinate system of the simulated to be 2D coordinate system)**;

modifying a position and a direction of viewpoint information of the three- dimensional CAD model of the inspection target object by using the coordinate transformation parameter **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**;

extracting a two-dimensional simulated image corresponding to the two- dimensional photographed image from the three-dimensional Computer Aided Design model after the viewpoint information is modified **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6, which teach extracting a 2D simulated image corresponding to the 2D real image from the 3D CAD model after the part corresponding to the reference parts of the inspection objection in the real image is recognized/identified)**; and

creating a defect image by fitting the two-dimensional photographed image including the defect to the two-dimensional simulated image, and depicting the defect image on the three-dimensional Computer Aided Design **(see, e.g., pars. 58-59 and 71 and FIGS. 1 and 4, which teach creating the adjusted defect image by adjusting the 2D defect image showing the defect detected by the detection unit to fit the 2D simulated image, and depicting the adjusted defect image on the 3D CD model).**

Shota teaches extracting, from the 3d CAD model, a simulated image that corresponds to the real image and contains the reference parts of the real image **(see, e.g., pars. 57 and 62-65 and FIGS. 1 and 4-6)**. The examiner believes that the cited portion of Shota may suggest estimating "a coordinate transformation parameter to transform a first coordinate system corresponding to the three-dimensional CAD model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image based on the shape recognized by the identification shape unit and on a three-dimensional CAD model of the inspection target object" and modifying "a position and a direction of viewpoint information of the three-dimensional CAD model of the inspection target object by using the coordinate transformation parameter."

However, for the interest of compact prosecution, the examiner relies on Sakamoto in the analogous art that explicitly teaches the quoted limitations. Sakamoto teaches:

a coordinate transformation parameter estimation unit to estimate a coordinate transformation parameter to transform a first coordinate system corresponding to the three-dimensional CAD model into a second coordinate system corresponding to a viewpoint of the imaging device that has captured the two-dimensional photographed image based on the shape recognized by the identification shape unit and on a three-dimensional CAD model of the inspection target object **(see, e.g., pars. 156-165 and FIGS. 2 and 5-7 of Sakamoto, which teach estimating the relative position and posture of the camera between the selected frame and the reference frame to transform a set of coordinates corresponding to the 3d model into another set of coordinates corresponding to the viewpoint of the camera that has captured the reference frame based on the 3d model and the 3d shape data therein)**;

a three-dimensional CAD model position change unit to modify a position and a direction of viewpoint information of the three-dimensional CAD model of the inspection target object by using the coordinate transformation parameter **(see, e.g., pars. 166-167 and FIG. 2 of Sakamoto, which teach modifying a position and posture of viewpoint information of the 3d model by associating the**

coordinates of the target image with the 3d model based on the position and posture of the camera that acquired the reference image and the condition, i.e., the internal parameter and the distortion correction parameter, for generating the 3d model).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the viewpoint of the 3d CAD model as taught by Sakamoto because doing so would yield predictable results of allowing an accurate estimation of the position of the defect when the viewpoints are modified **(see, e.g., par. 166 of Sakamoto and MPEP 2143(I)(D)).**

20. Claim(s) 7 is/are rejected under 35 U.S.C. 103 as being unpatentable over Shota in view of Sakamoto and further in view of us patent application publication no. 2022/0383456 to Gulati et al. (hereinafter Gulati).

21. For claim 7, while Shota in view of Sakamoto does not explicitly teach, Gulati in the analogous art teaches correcting the coordinate transformation parameter through noise removal using machine learning **(see, e.g., pars. 62-64 of Gulati, which teach performing a denoise function using a deep learning model before performing image transformation)**.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Shota in view of Sakamoto to denoise before transforming as taught by Gulati because doing so would yield predictable results of allowing providing a better quality transformed image **(see, e.g., par. 63 of Gulati and MPEP 2143(I)(D)).**

Conclusion

22. **THIS ACTION IS MADE FINAL.** Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

23. Any inquiry concerning this communication or earlier communications from the examiner should be directed to WOO RHIM whose telephone number is (571)272-6560. The examiner can normally be reached Mon - Fri 9:30 am - 6:00 pm et.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Henok Shiferaw can be reached at 571-272-4637. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/WOO C RHIM/
Examiner, Art Unit 2676
/CHAN S PARK/
Supervisory Patent Examiner, Art Unit 2669